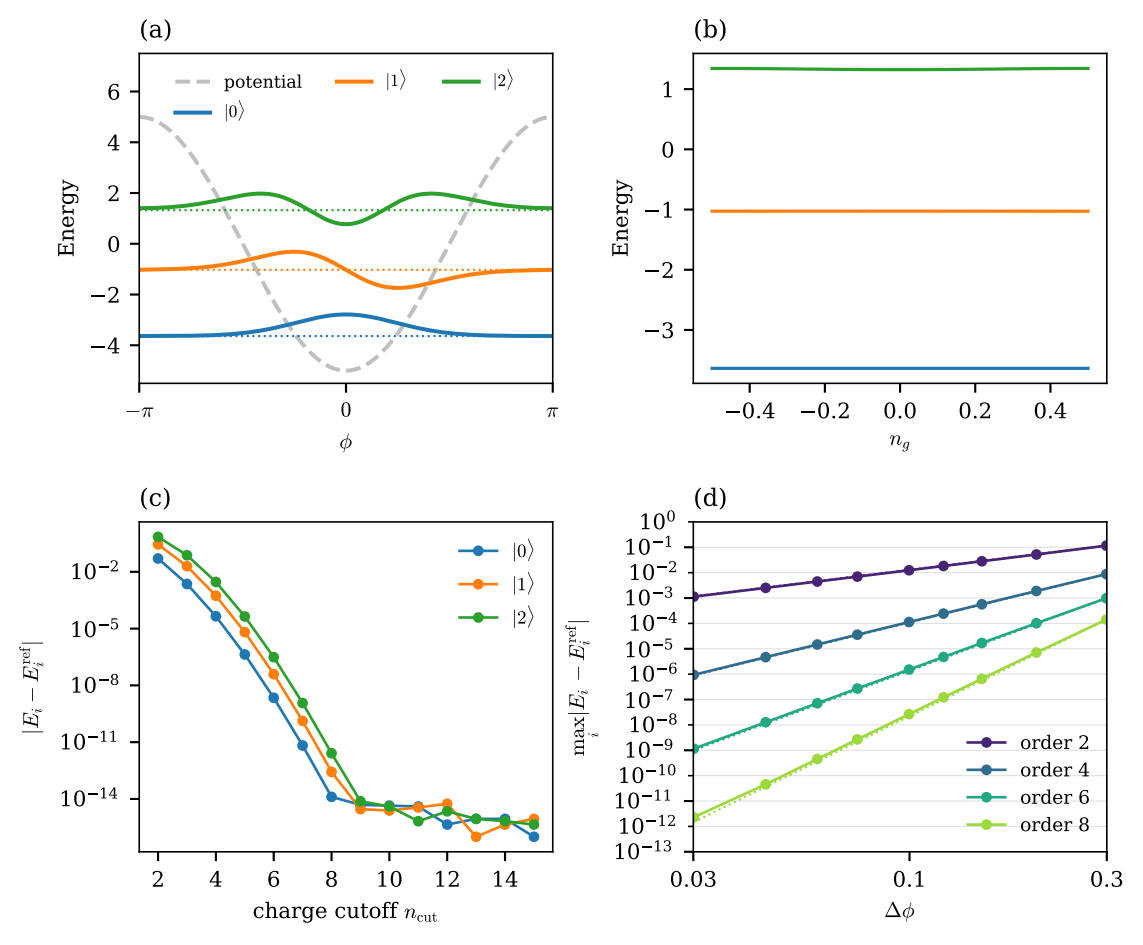
(a)
potential
$|1\rangle$
$|2\rangle$
$|0\rangle$
Energy
6
4
2
0
−2
−4
$-\pi$
0
π
ϕ
(b)
Energy
1
0
−1
−2
−3
−0.4
−0.2
0.0
0.2
0.4
n_g
(c)
$|E_i - E_i^{\mathrm{ref}}|$
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
$|0\rangle$
$|1\rangle$
$|2\rangle$
2
4
6
8
10
12
14
charge cutoff n_{cut}
(d)
$\max_i |E_i - E_i^{\mathrm{ref}}|$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
order 2
order 4
order 6
order 8
0.03
0.1
0.3
$\Delta\phi$